Proceedings of the 6th International Conference on Electronics and Sustainable Communication Systems (ICESC-2025)
DVD Part Number: CFP25V66-DVD; ISBN: 979-8-3315-5502-3

Development of a Speech-to-Text Emotion and Speaker Recognition System for Individuals with Hearing Impairments

1st Mrityunjay Gupta
Student, Jayshree Periwal International School
Jaipur, India
mrityunjay.coder@gmail.com

2nd Vinay Vishwakarma@omotec.in
Mentor, On My Own Technology
Mumbai, India
vinay.vishwakarma@omotec.in

***Abstract*—Communicative challenges experienced by individuals with hearing impairments may prove serious in attempts to recognize emotional aspects of spoken communication. The conventional speech-to-text systems are not applicable to the emotional aspects of speech. This results in less communication and miscommunication. Moreover, systems used previously may not work well with different speakers and speaking styles, hence causing further barriers to communication to the hearing impaired community. A Speech-to-Text Emotion and Speaker Recognition System was developed that merges accurate speech transcription with emotion and speaker detection in order to arrest this problem. This approach involves building of audio datasets containing four classes (A1, A2, A3, A4), followed by extracting the main feature representation using MFCCs. A CNN model was therefore utilized and was able to achieve an accuracy of 99.3% during the audio classification. Thus, speakers from varied backgrounds were assured of accurate transcriptions. To achieve emotion recognition, the pre-trained Wav2Vec2 model serves to classify emotions into neutral, happy, sad, angry, and fearful categories having attained high degrees of accuracy. Performance tests with hearing-impaired subjects included highly accurate transcriptions and emotion detection. As it gave a real-time interface, responses delivered quite smoothly, averaging around 250 ms response time, preventing hampering communication. Further, scenario-based evaluations used for different profiles of speakers validated the capability of the system to accommodate different voices and speaking styles. These results all approve the system's ability to bring the hearing impaired closer to the information, most commonly enabling accurate speech-to-text conversion, emotional insights, and speaker recognition. This research significantly adds to assistive technologies since it broadens channels of communication, all in favor of the hearing impaired.**

Keywords—Speech-to-text, Emotion Recognition, Assistive Technology, Deep Learning, Speaker Audio Classification.

I. INTRODUCTION

Hearing impairment profoundly affects communication, leading to significant social isolation and emotional distress among individuals. However, this is far from the possibility of achieving optimal specific understanding since those with hearing loss grasp emotions from their surroundings using other ways, primarily using visual cues. There are several challenges that people suffering from hearing impairment have to face. The most significant problem is a lack of ability to understand the spoken word itself, which becomes even more difficult in noisy environments, where the dazzle of words and movements in the lips or faces of others speaking [1]. Loudspeakers are of no use under such circumstances, although they are frequently used in public areas as open-space project-management speakers. In contrast, this ability creates the hearing-impaired person shut out and marginalized ([UCSF] Health) [2]. Cochlear implants do make speech partly perceivable to certain people, even some with complete loss of hearing, but with crucial considerations. Most of them still depend on visual modes like lip reading and sign language. Thus the efficiency of communication can well be severely diminished by other elements than these, principally poor light and noise in the background (Cincinnati Children's) [3]. The existing systems did pioneer a great advancement in communication for those who are impaired in hearing, by integrating spoken languages and the use of sign languages, which are further translated into texts. Furthermore, the invention of these systems is based on deep learning and machine learning, where a speech recognition engine converts an incoming speech source into text input, which is then read into the engine and translated further into sign language which can be seen on an onscreen display. Technically, these have advanced in improving rehabilitation, but mere sound has never been a means cooperating enough for fulfillment or closure of voids in social interactions. Rather, this requires full availability for emotional understanding or recognized context within the perception, along with the speaker's identity. The proposed system called *AERIS*, **A**udio-based **E**motion and **R**ecognition of **I**ndividual **S**peakers, aims to bridge this gap and bring the novel impulse within the user's communication. *AERIS* combines speech-to-text conversion, emotion recognition, and speaker identification into one coherent platform to obtain better communication experiences in this manner. Advanced machine learning allows real-time transcription of speech into text and analysis of sound signals in both emotional classification and speaker recognition. The Google Speech-to-Text API [23] serves for efficient and accurate speech-to-text conversion. Emotion recognition is based on the pretrained Wav2Vec 2.0 [22] model with deep

learning for feature extraction and classification. Speaker identification is done by using Convolutional Neural Networks (CNNs) [21] to extract unique speaker embeddings that distinctly identifies different speakers and voices. This not only helps individuals, but it potentially changes the way communication functions in many industries. The innovation of enhanced communication brings an advantage to individuals as well as entire industries. Improving inclusiveness in communication spaces enhances the quality of life, be it in individual or professional spheres such as healthcare and education.

II. LITERATURE REVIEW

The research by **Amandeep Singh Dhanjal et al.[4]** aimed at the improvement of communication for people with impaired hearing using diverse assistive technologies, classifying them into Assistive Listening Devices (ALD), Augmentative and Alternative Communication (AAC), and alert systems. Existing tools were reviewed under some parameters such as speech enhancement, recognition of sign language, application of learning, telecommunications systems, and text speech. The results showed that many innovative prototypes were available, yet the majority of them were not on the market and, therefore, have not gone a long way in being used by the target audience. This research shows how these devices have reduced communication barriers and have improved educational and social experiences for people with hearing impairment. Such technologies need further development and eventually become commercialized to bring them under the scope of inclusion and improve the quality of lives of this population and could be a strong foundation for future impacts in assistive technology.

The study by **Ajay K. J et al.[5]** revealed a person's daily life experiencing hardship due to hearing impairment and the effectiveness of Ear Assist through empirical proof and user testimonies. Ear Assist (EA), smart eyeglasses, were proposed to positively identify the communication barrier and increase accessibility for over 360 million individuals worldwide with hearing impairment. The evidence shows the potential of instruments and aid such as Ear Assist in bringing possible contributions to inclusiveness and social equity. The study clearly portrays how technology can improve the lives of people with hearing problems. The foundation for further works in assistive technologies to promote better communication and a more inclusive environment for people with hearing impairments has been set.

Recent technological advancements have dramatically improved all computer vision (CV) applications and subjected its quality of life to factory automation, self-driving cars, surveillance, healthcare, and assistive technology (AT). The study by **K. C. Shahira et al.[6]** presented a summary of different types of assistive technology meant for people suffering from vision, hearing, and verbal disabilities. It discussed the need for the integration of CV innovations into assistive technologies to solve the unique challenges of these populations. Findings can again recognize that further development in AT will lead to greater accessibility and inclusion for persons with disabilities, thus bettering quality of their life and independence.

The research by **Virkkunen et al.[7]** focused on an assistive mobile application that would use augmented reality and automatic speech recognition technologies to help individuals with hearing loss understand verbal communication. The mobile augmented reality with video-based augmentations integrated automatic speech recognition using state-of-the-art neural networks. The transcriptions from the speech recognizer are displayed in speech bubbles overlaid on the augmented reality view of the conversation partner. Thus, the distance between the speaker and transcriptions is minimized. User tests conducted with hearing-impaired participants yield an effectiveness validation of the application, finding them aided by the augmented reality interface in pursuing conversations. However, improvements are desired for visual separation and identification of speakers in group conversations and additional accuracy in recognizing speech. These findings show the promise that the conjoined field of augmented reality and speech recognition could provide individuals with hearing impairments and give cause for further research developments in assistive technologies.

The study by **Beskow et al.[8]** proposed speech and audio signals for visualizations to strengthen and develop communication improvements for those people who would differ from mainstream populations, notably hearing-impaired individuals. This has been hinged on their recognition that acoustic information needs supplement or replacement by visual cues and the extent to which audiovisual modalities are considered natural complements. The same visualizing techniques were suggested and demonstrated for everyday applications in the lives of hearing-impaired persons. Applications were given as examples to imagine how these methods would cater to a more growing population, including children, middle-aged, and older individuals, all of whom suffer hearing loss. Such understanding shows how important visualization of sound and speech could prove to be in promoting understanding and participation in social contexts. The result underlines the importance of visual cues, making it a valuable research contribution toward the promotion of assistive technologies by which societies may communicate more easily and interact better with hearing-impaired people.

The study by **Beskow et al.[9]** aimed to enhance communication for the hearing impaired through various methods of visualization for speech and audio signals. Since acoustic information usually requires extra or alternative sources of visual cues. This research had to focus on the natural complementarity of audiovisual modality as, rather than employing visual cues, complementing the verbal expression with spatial realization was thought a possible kind of mediation. Therefore, several visualizations have been proposed, with examples that will show how these mechanisms can stimulate equal participation in this aspect of everyday life for a wide range of users, children, middle-aged people and older adults suffering from hearing impairment. Visualization techniques were able to demonstrate how well they work through applications that help such users understand vastly better what is said around them and what sounds simulate the environment. Findings revealed that promising representations

of visual communication in speech and sound should encourage and facilitate socialization and participation.

This study by **Mary Jane C. Samonte [10]** aimed to create an e-tutor system that could help students having any kind of physical disability and learning under really tough circumstances. The new system had provided all the tools to the teacher in order to teach the lessons and conduct assessments through a multi-approach platform. With this system, students could now avail their learning modules through home usage. The earlier emphasis of this research was on statistics, which was found to have a sheer base for further advanced learning in big data and analytics. This system, using assistive technologies like speech-to-text conversion, gamified learning, and handwritten character recognition, aimed to teach basic statistics to Filipino students through Filipino Sign Language (FSL). The implementation of interactive and engaging experiences through four modules made for inclusivity in education.

The study by **Mandlenkosi Shezi et al.[11]** sought to investigate the challenges encountered by the deaf people when talking, especially with the current mobile applications using Speech to Text technology. Most of these applications were found to be inconsistent and were not all-inclusive of different forms of hearing impairments. Then a model using Multiple-Speaker Classification was conceptually introduced in order to improve the communication. A prototype of a mobile application, Deaf Chat, has been built from it. Evaluation surveys have shown high acceptance from users and that Deaf Chat meets the needs of the hearing impaired well enough that it could enhance their conversational experience and improve social interaction.

The study by **K. C. Shahira et al.[12]** focused on the transformational miracles of new-age computer vision technology towards improving the lives of disabled people. Applications developed using increased processing power, machine-learning techniques, or deep learning techniques are prevalent in various fields such as factory automation, health care, and assistive technologies. Under the scope of acknowledging reasonable social commitment to AT technology, which includes services and products designed to aid disabled persons, they summarized several AT methods for various disabilities, such as vision, hearing, and verbal impairments. This overview provides insight into the vast potential of the innovations in computer vision towards truly revolutionary changes in making things more accessible and supportive of people with disabilities.

The study proposed by **N. Palanivel et al.[13]** developed an Automated Visual Recognition System (AVRS) that has been planned as an enhancement lip-to-speech converter system for the user which will provide an effective medium to the hearing-impaired individual. The system used the webcam for capturing lip movements of the speaker, which are processed in real-time with a microcontroller AVR. The lip movements are grabbed through image processing techniques and then converted to speech using the text-to-speech(TTS) algorithm. The system has had the synthesized speech outputted through the speaker and hence could be used without lip reading and sign language. Test runs have shown a promising accuracy and efficacy when used by people with difficulties in speech, making it an affordable portable solution for improved communication-in-health as well as education purposes in divergence settings.

The study by **E. Saranya et al.[14]** presented a new technique for an interaction between normal-hearing people and hearing impaired people and addresses limitations in current assistive devices like hearing aids or cochlear implants-particularly they are not very much effective in a noisy environment. State-of-art speech recognition technology has been applied in the system by factoring the noise environment into its signal processing features; nevertheless, it is done without having a physical engagement with the user. The system essentially constitutes a portable device that houses an acoustic beamformer for directional noise reduction and is capable of speech to text conversion using keyword spotting algorithms. Tests conducted on this system have indicated a positive outcome in realizing bidirectional communication and thus have the potential to be part of a solution for facilitating the improvement of multi-auditory environments. Essentially, this technique sought to empower hearing-impaired people in better communicating in highly challenging environments.

The research by **Vidharsh Chuckun et al.[15]** developed HandyApps, a kind of mobile application that aims to advance life quality for people with disabilities, which includes those who are visual, speech, and hearing impaired. The application contained tools such as face recognition, object recognition, text reading recognition, and face recognition to assist visually impaired users. And for people who are hearing impaired, it has the Speech-to-Text functions as well as a sign language application based on the sign language with alphabets and numbers. This application also serves as a Text-to-Speech with some preset phrases intended for speech-impaired users. The tests with such users revealed very positive feedback about the usefulness of the app, confirming its promise toward self-sufficiency with respect to everyday life and accessibility.

The study by **Ohene-Djan et al.[16]** addressed the inability to understand the underlying emotions, intonations and volume in conventional subtitles, therefore reducing the ability to grasp media for deaf and hard-of-hearing individuals. The emotional subtitles system used formatting methods, like colour, font type and size, and annotations, to convey the emotions in the speech visually. A structured subtitle specification system using XML-based formatting and a Flash-based movie player was used, allowing for a dynamic representation of the emotions in the speech. It involved the usage of different colours for different characters to further improve understanding. The results showed that on one hand, emotion subtitling is a better version than traditional ways, further research is required for making the formatting methods more effective, and testing its utility across varied demographics, especially in educational and entertaining content for the welfare hearing-impaired people.

Dua et al.[17] addressed the roadblocks in the conversations between hearing-impaired and the hearing group by bringing forth a speech recognition tool that works by translating continuous speech into words and images, through the use of Automatic Speech Recognition (ASR) along with Mel Frequency Cepstral Coefficients (MFCC) and Vector

Quantization (VQ) to extract speech features. The features were then paired with the pre-installed templates. Two primary modules, which are Speech-to-Text (STT) conversion and an image retrieval system were used to improve the understanding of the hearing-impaired. Tests on a vocabulary of 20 day-to-day spoken phrases revealed a recognition accuracy of 95%, when it was not suited for a specific speaker. Yet, a difficulty that exists is scalability, as a greater range of vocabulary tremendously changes the recognition rates. The results revealed the need for a bigger database of several languages along with real-time adaptability for the betterment in inclusivity of hearing-impaired individuals.

Nagels et al.[18] focused on the evolution of speech emotion identification in normal-hearing (NH) and hearing-impaired children, attempting to bridge the gap in the presence of high-quality audio tests to evaluate how the NH and cochlear implant (CI) users perceive the emotions in the audios. The EmoHI test that uses pseudo speech stimuli was proposed, which is meaningless phrases spoken with different emotional tones. NH Dutch and English children along with prelingually deaf Dutch children with CIs were assessed to analyse trends in performance in relation to age. Results revealed that for the NH children, accuracy increased with age, yet failing to reach the level of accuracy by 10-12 years that adults have. CIs, however, had highly variable results, with some having the same results as their NH peers, while others performing at chance level, which highlighted that factors like auditory experience and interpersonal and emotional growth, played a part in their capability to identify emotions, rather than just age. The study showed that EmoHI is a feasible system to evaluate how well people identify emotion from speech, though further research is required to make the best use of this system over different linguistic and hearing-impaired groups.

III. METHODOLOGY

A. Data Collection and Preprocessing

The dataset consists of audio files categorized into A1, A2, A3, and A4 inclusive of four classes of speech samples in .wav form. The source of audio files was audiobooks [26] and other sites to capture varying speech signals. The speech samples were processed using Speechify AI [27] for changing the samples into different voice types such as old man, old woman, young man, and young woman. The path to all audio was extracted, and the associated labels tagged onto each file. Such label file information is stored in a CSV file that serves as the basis for further analysis. The audio files were imported using librosa, a popular library known to have been built for audio processing tasks [19]. As part of the first phases toward exploring the dataset, we visualized the raw signals in forms of their waveforms as well as spectrograms allowing insight into the structure of the data. The representation of an audio signal over time is illustrated by the waveform, which shows how an audio signal would be perfectly created in its temporal properties. The content of the frequencies for the audio signal over time is also plotted in the spectrogram representation, providing valuable information on the signal's spectral features. The visualizations help to understand the distinctions in the different classes in the dataset and serve as a preliminary study prior to more detailed feature extraction. Examples of the waveform as shown in Fig 1, Fig3 ,Fig 5, Fig 7 and spectrogram for each class (A1, A2, A3, A4) are shown in Fig 2, Fig 4, Fig 6 and Fig 8, capturing the specific nature of the audio signals in each category. The preprocessing of data ensured that the model received input in a suitable format.

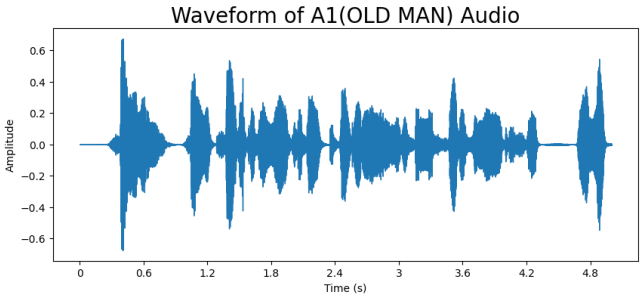


Fig 1.Waveform of A1 (Old Man) Audio

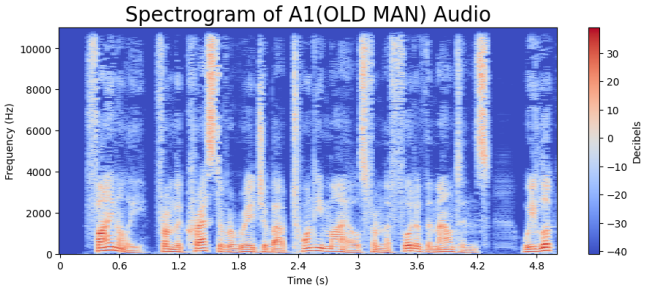


Fig 2. Spectrogram of A1 (Old Man) Audio

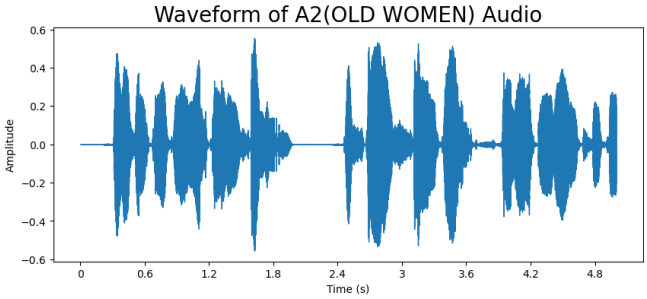


Fig 3. Waveform of A2 (Old Women) Audio

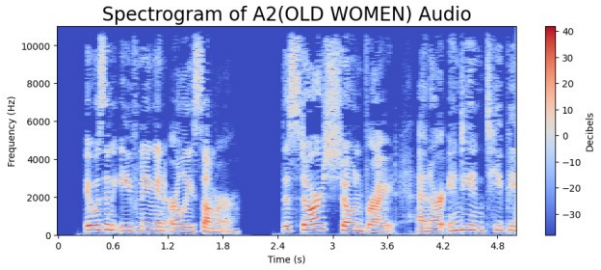


Fig 4. Spectrogram of A2 (Old Women) Audio

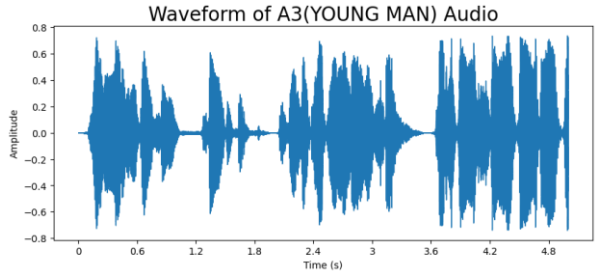


Fig 5. Waveform of A3 (Young Man) Audio

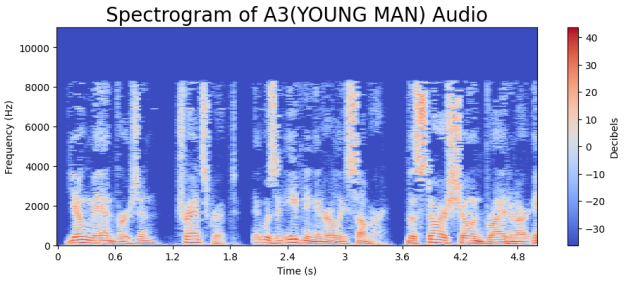


Fig 6. Spectrogram of A3 (Young Man) Audio

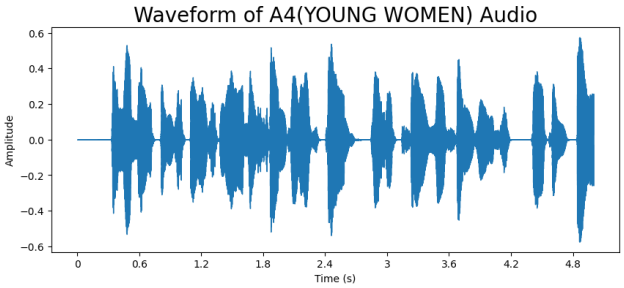


Fig 7. Waveform of A4 (YoungWomen) Audio

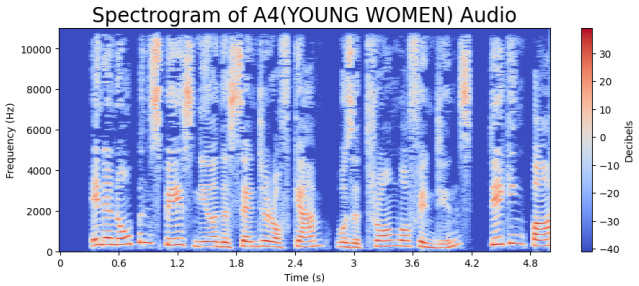


Fig 8. Spectrogram of A4 (Young Women) Audio

B. Feature Extraction

Extracting features from the raw audio files involves using Mel-frequency cepstral coefficients (MFCC), which is a standard feature representation for audio classification tasks [20]. The MFCC captures the frequency content of an audio signal and is used most prominently for tasks such as speech and emotion recognition. The MFCC extraction process loads each audio file with fixed duration and offset, followed by MFCC feature computation. The MFCC feature vectors are averaged through time, resulting in a 40-dimensional feature vector for each audio file, which is important for training the machine learning model. The MFCC features are extracted from all audio files in the dataset, thus forming a feature matrix, X_mfcc, where each entry corresponds to the MFCC vector of a given audio file.

C. Model Architecture

The architecture of AERIS's CNN classifier [21] is designed for audio and speaker classification, specifically spectrograms of speech or emotional expressions; It extracts features from the audio input Mel-frequency cepstral coefficients (MFCCs). This model consists of fully connected layers, convolutional layers and max pooling layers, used for the accurate classification of audio samples into four distinct classes. **Input Layer-** The input into the model contains a sequence of 40 MFCC features corresponding to each audio sample, structured as a 1D array with dimensions (40, 1). These 40 features are compressed spectral measures of the audio signal and are sent to convolutional layers for further processing. **Convolutional and Max-Pooled Layers:** This architecture contains three 1D convolutional layers, each followed by max-pooling. These are designed to capture hierarchical patterns in the temporal domain for the given input audio signal. **First Convolutional Layer:** It has 64 filters and kernel sizes of 3, applying an activation function of ReLU that introduces non-linearity and facilitates the network to learn complicated features [24]. The output of this layer is passed through the max-pooling layer having a pool size of 2. This downsampled the feature map while retaining the most important elements of it. **Second**

Convolutional Layer: The same procedure as the first layer is followed, with 64 filters, a kernel size of 3, and ReLU activation. Features from the first convolutional layer are further refined from the second onward, and output is again downsampled through another max-pooling operation. **Third Convolutional Layer:** This layer is just like the previous two. Further processing of the audio features captures high-level patterns. **Flatten Layer:** After convolutional layers and max-pooling, the feature maps are flattened into a 1D vector. This results in preparing the data for the fully-connected layers in which each feature will individually be treated as an input into the dense neurons. **Fully Connected (Dense) Layers:** The output flattened from the above step is forwarded through two fully connected layers. The first dense layer consists of 512 neurons and has the ReLU activation function. The second dense layer is the output layer that contains 4 neurons to correspond with the 4 classes (for instance, A1, A2, A3, A4). This layer uses a softmax activation to calculate class probabilities, and the final output reflects the chances of that input sample belonging to each class[25]. The softmax function plays an important role in normalizing the output values into a probability distribution.

D. Model Compilation and Training

The model is compiled using categorical cross-entropy loss which is suitable for multi-class classification tasks , and the use of stochastic gradient descent (SGD) [28] optimizes the performance by adjusting the weights of the model in minimizing the loss calculated from standard backpropagation. During training, accuracy is used to evaluate the performance of the model. The model was allowed to train on the training dataset for 100 epochs with an early stopping callback with the patience of 20 epochs[29], and generalization capability was checked on the validation dataset. The network uses ReLU activation throughout, promoting learning by introducing nonlinearity, while providing proper multi-class classification through softmax activation in the output layer.

E. Evaluation

The evaluation phase was mainly designed to assess the performance of the CNN-based speaker recognition system and the pretrained emotion classification model with unseen data. The analysis of the training history of the CNN model involved the tracking of the training and validation accuracy and loss across epochs that were intended to provide a good view of the learning behavior of the model in trend terms from overfitting, or underfitting symptoms. The extracted MFCC features were used for making the predictions on the unseen audio samples, and the most probable class label was given for the highest predicted probability. Wav2Vec2.0 was a pretrained model by which the performance of emotion detection was assessed compared to its predicted emotion labels and matching confidence values.

F. Emotion Classification Model

For emotion classification, a pre-trained deep learning model is used with the help of transformers library, specifically superb/wav2vec2-base-superb-er [22] to classify the speaker's emotional state based on recorded audio. This model is a specialized instance of the Wav2Vec2 architecture, which uses self-supervised learning for meaningful direct speech representations from raw audio waveforms. The recorded audio is processed accurately into a format suited to the model, which is then passed as input for feature extraction and classification tasks. Based on the patterns of the speaker's vocal the model assigns the calculated probability scores to a number of emotion categories, including neutral, happy, sad, angry, and fearful. The emotion label represented by the highest probability will be predicted as an emotional state [30].

G. Speech to Text Conversion

The audio would be converted through an advanced process such that they could be transformed into written formats using a technique called Google's Speech Recognition API [23] : automated sound absorption-from intensive sound analysis and text matched with its model of possible sounds. This starts with the audio-capturing microphone, preprocessing it that serves the purposes of clarity and noise reduction. The sound file transforms to become spectrographic representations of the audio frequencies, where the phonemes are identified as the smallest non-composite units in the sound. Including deep learning models within AERIS is a way to apply context and language models to the task and achieve a high transcription accuracy-efficiency tradeoff. Thus, this not only makes the speech-to-text possible in real time but also provides capabilities of different languages and accents in their delivery, proving to be an asset in accessibility to communication, particularly for the hearing impaired.

IV. Results

An evaluation of AERIS was performed through controlled conditions for accuracy in transcribing, emotion recognition, and real-time feedback efficiency. The CNN Classifier for speaker recognition was directly tested with audio samples of A1, A2, A3, and A4, achieving an overall accuracy of 99.3%, as shown in **Fig 10**. **Fig 10** and **Fig 11**. illustrate the model's accuracy and loss trends over 100 epochs. Training and validation accuracy (shown in Fig 10) gradually increases and stabilizes at around 100%, indicating effective learning . The loss decrease shown in Fig 11. was sharp during the initial epochs to near-zero values, affirming convergence. The closeness of training and validation metrics demonstrates the model's robustness and ability to generalize. The emotion classification model, based on a pretrained model utilizing Wav2Vec2, was appraised on a dataset containing samples of neutral, happy, sad, angry, and fearful speech, achieving reasonably good accuracy. Trial with user groups (hearing-impaired people) of diverse backgrounds illustrated the definite

efficacy of the system because different vibration patterns representing emotions in speech enabled more efficient communication, in effect, with an average response time of 250ms ensuring real-time interaction with no perceivable delay. The confusion matrix as shown in Fig 9. Illustrates how well the model classifies its inputs into four categories: Old Man (A1), Old Woman (A2), Young Man (A3), and Young Woman (A4). High diagonal values mean accuracy while correct predictions justify with true labels above. Scenario-based testing **Fig 12a**, **Fig 12b**, **Fig 12c**, and **Fig 12d** used different speaker profiles (Old Man - A1, Old Woman - A2, Young Man - A3, and Young Woman - A4) to prove the robustness of the system across varying voices, ages, and speech styles.

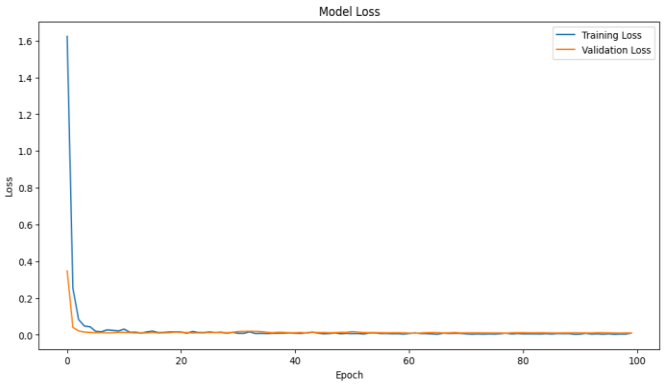


Fig 11. Evaluation of Training and Validation Loss Graph

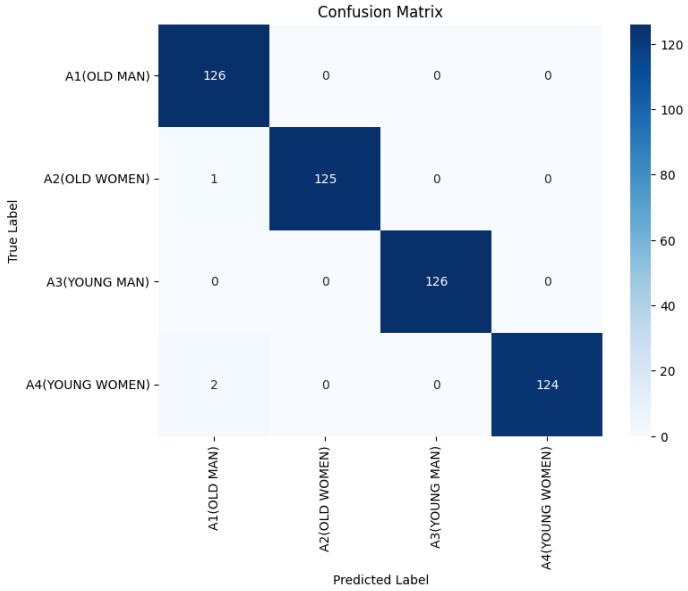


Fig 9 . Results Evaluation using Confusion Matrix

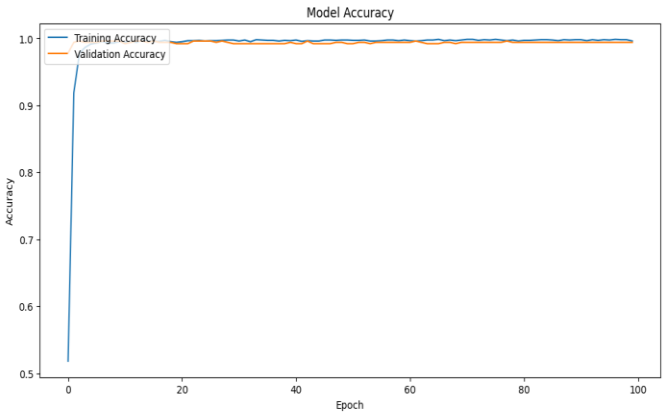


Fig 10 . Evaluation of Training and Validation Accuracy Graph

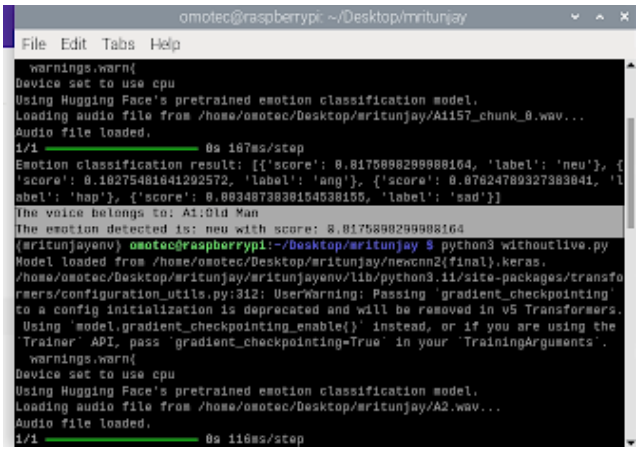


Fig 12a. Test Scenario 1 (Old Man - Class A1)

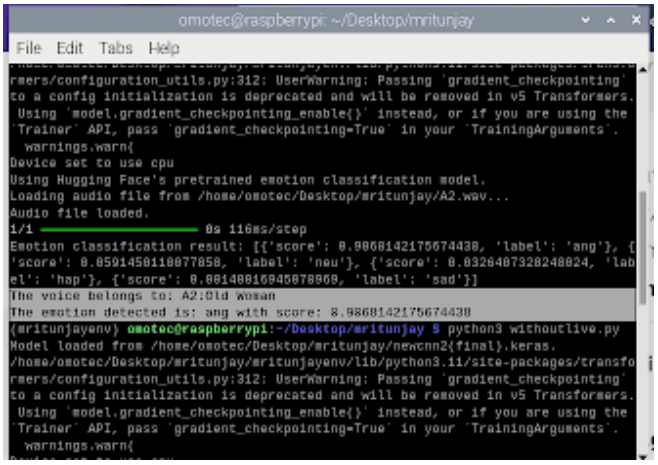


Fig 12b. Test Scenario 2 (Old Woman - Class A2)

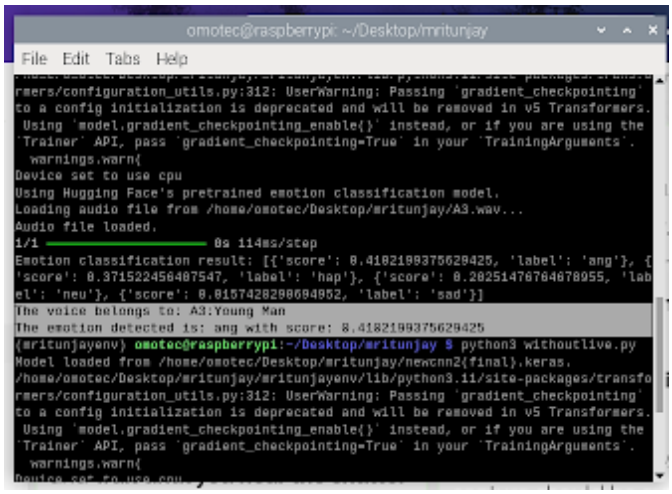


Fig 12c. Test Scenario 3 (Young Man - Class A3)

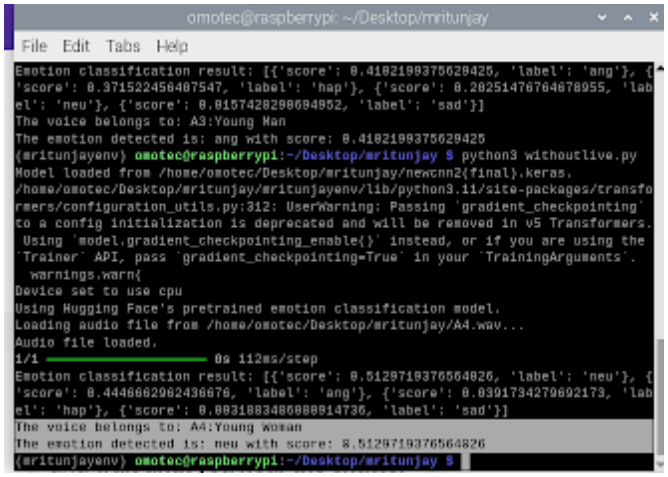


Fig 12d. Test Scenario 4 (Young Woman - Class A4)

V. CONCLUSION

The research introduces an innovative **Speech-to-Text Emotion and Speaker Recognition System, AERIS** engendered to augment the communication ability of hearing impairment people. The integration of state-of-the-art deep learning technologies and advanced audio signal processing terms allows effective speech-to-text conversion ,emotion recognition and speaker recognition . Thus, AERIS fulfills an important gap in the assistance technology by not only generating speech but also understanding individual speakers which is very important in multi-speaker environments. The experimental validation has shown the entire robustness of the system. **The CNN-based speaker recognition model attains an accuracy of 99.3%,** understandably distinguishing between different speakers thus ensuring personalized transcription accuracy. The emotion classifier, **using the pre-trained Wav2Vec2 framework**, also performed very well in prediction, classifying the emotional expressions. The real-time testing with hearing-impaired users further proved the practical usability of AERIS, rendering the communication more natural and emotionally intelligent. But besides its direct value as an assistive technology, this research goes a long way toward demonstrating how deep learning-enabled multimodal communication systems can find application in assistive technologies. The results of the study suggest outlooks for application in the healthcare domain, the educational domain, and the domain of human-computer interaction, where emotional intelligence is important for an effective engagement.

Future works will focus on incorporating multi-modality in the form of feedback such as haptic feedback or visual cues. This work acts as a milestone in the domain of speech recognition and non-verbal communication and helps set the tone for a more inclusive and context-aware assistive communication system that acts to promote greater social and technological inclusion for people with hearing impairments.

ACKNOWLEDGMENT

I would like to express my gratitude to the mentors Ms. Sanskruti Chavan and Mr. Radhakrishnan Rajan of On My Own Technology Pvt. Ltd. for extending their help in carrying out the research.

REFERENCES

[1] Kumar, Arun, and Arun Kumar. “The Challenges Faced by People With Hearing Impairment/Deafness! - by Dr. Arun Kumar | Lybrate.” Lybrate, 12 Aug. 2019, www.lybrate.com/topic/the-challenges-faced-by-people-with-hearing-impairment-deafness/c560d9dad45758f31a28a08a93551338.

[2] “Communicating With People With Hearing Loss.” ucsfhealth.org, 21 May 2024, www.ucsfhealth.org/education/communicating-with-people-with-hearing-loss.

[3] Hearing Loss Affect on Communication | Advice for Parents. www.cincinnatichildrens.org/health/h/hearing-communication.

[4] Dhanjal, Amandeep Singh, and Williamjeet Singh. "Tools and techniques of assistive technology for hearing impaired people." 2019 International conference on machine learning, big data, cloud and parallel computing (COMITCon). IEEE, 2019.

[5] Xavier, Alvis, et al. "Ear Assist for Hearing Impaired." 2024 International Conference on Futuristic Technologies in Control Systems & Renewable Energy (ICFCR). IEEE, 2024.

[6] Kbar, Ghassan, et al. "Assistive technologies for hearing, and speaking impaired people: a survey." Disability and Rehabilitation: Assistive Technology 12.1 (2017): 3-20.

[7] Virkkunen, Anja. "Automatic speech recognition for the hearing impaired in an augmented reality application." (2018).

[8] Beskow, Jonas, et al. "Visualization of speech and audio for hearing impaired persons." Technology and Disability 20.2 (2008): 97-107.

[9] Beskow, Jonas, et al. "Visualization of speech and audio for hearing impaired persons." Technology and Disability 20.2 (2008): 97-107.

[10] Samonte, Mary Jane C. "An assistive technology using fsl, speech recognition, gamification and online handwritten character recognition in learning statistics for students with hearing and speech impairment." Proceedings of the 6th International Conference on Frontiers of Educational Technologies. 2020.

[11] Shezi, Mandlenkosi, and Abejide Ade-Ibijola. "Deaf chat: A speech-to-text communication aid for hearing deficiency." Advances in Science, Technology and Engineering Systems Journal 5.5 (2020): 826-833.

[12] Shahira, K. C., C. J. Sruthi, and A. Lijiya. "Assistive technologies for visual, hearing, and speech impairments: Machine learning and deep learning solutions." Fundamentals and Methods of Machine and Deep Learning: Algorithms, Tools and Applications (2022): 397-423.

[13] Palanivel, N., et al. "Visual recognition system for hearing impairment using internet of things." AIP Conference Proceedings. Vol. 3075. No. 1. AIP Publishing, 2024.
[14] Saranya, E., B. Baron Sam, and R. Sethuraman. "Speech to text user assistive agent system for impaired person." 2017 IEEE International Conference on Smart Technologies and Management for Computing, Communication, Controls, Energy and Materials (ICSTM). IEEE, 2017.
[15] Chuckun, Vidharsh, Ganeshwaree Coonjan, and Leckraj Nagowah. "Enabling the Disabled using mHealth." 2019 Conference on Next Generation Computing Applications (NextComp). IEEE, 2019.
[16] Ohene-Djan, James, Jenny Wright, and Kirsty Combie-Smith. "Emotional Subtitles: A System and Potential Applications for Deaf and Hearing Impaired People." CVHI. 2007.
[17] Dua, Tanvi, et al. "Speech recognition technology for hearing disabled community." International Journal of Advanced Computer Research 4.3 (2014): 882.
[18] Nagels, Leanne, et al. "Development of vocal emotion recognition in school-age children: The EmoHI test for hearing-impaired populations." *PeerJ* 8 (2020): e8773.
[19] McFee, Brian, et al. "librosa: Audio and music signal analysis in python." *SciPy* 2015 (2015): 18-24.
[20] Charisma, Atik, M. Reza Hidayat, and Yuda Bakti Zainal. "Speaker recognition using mel-frequency cepstrum coefficients and sum square error." *2017 3rd International Conference on Wireless and Telematics (ICWT)*. IEEE, 2017.
[21] Chaturvedi, Abhay, et al. "Classification of sound using convolutional neural networks." *2022 5th International Conference on Contemporary Computing and Informatics (IC3I)*. IEEE, 2022.
[22] Pepino, Leonardo, Pablo Riera, and Luciana Ferrer. "Emotion recognition from speech using wav2vec 2.0 embeddings." *arXiv preprint arXiv:2104.03502* (2021).
[23] Anggraini, Nenny, et al. "Speech recognition application for the speech impaired using the android-based google cloud speech API." *TELKOMNIKA (Telecommunication Computing Electronics and Control)* 16.6 (2018): 2733-2739.
[24] Banerjee, Chaity, Tathagata Mukherjee, and Eduardo Pasiliao Jr. "An empirical study on generalizations of the ReLU activation function." Proceedings of the 2019 ACM Southeast Conference. 2019.
[25] Sharma, Sagar, Simone Sharma, and Anidhya Athaiya. "Activation functions in neural networks." *Towards Data Sci* 6.12 (2017): 310-316.
[26] ---. “Audible.” Audible.com, www.audible.in/ep/free-listens.
[27] “---.” Speechify, speechify.com/?srsltid=AfmBOoqeiT8C-4QfVp-tUnVRrrz0XttbwBN90RKo7i_0csE07As9pj_3.
[28] Tian, Yingjie, Yuqi Zhang, and Haibin Zhang. "Recent advances in stochastic gradient descent in deep learning." *Mathematics* 11.3 (2023): 682.
[29] Hussein, Bootan M., and Shareef M. Shareef. "An empirical study on the correlation between early stopping patience and epochs in deep learning." *ITM Web of Conferences*. Vol. 64. EDP Sciences, 2024.
[30] Singh, Jagjeet, Lakshmi Babu Saheer, and Oliver Faust. "Speech emotion recognition using attention model." *International Journal of Environmental Research and Public Health* 20.6 (2023): 5140.